HW1: QFT I

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1. Lattice and Paths

For a d-dimensional lattice, we define

$$x^\mu = n^\mu \Delta, \qquad n^\mu \in \mathbb{Z}^d. \tag{1}$$

For example, in two dimensions there are two independent lattice directions,

$$\hat{e}_1, \qquad \hat{e}_2, \tag{2}$$

and in three dimensions,

$$\hat{e}_1, \qquad \hat{e}_2, \qquad \hat{e}_3. \tag{3}$$

In general, for four dimensions,

$$\hat{e}_i, \qquad i = 1, 2, 3, 4. \tag{4}$$

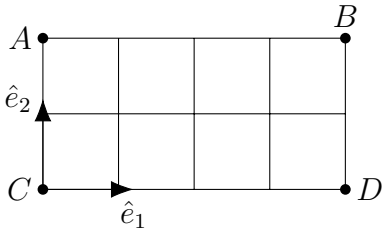

Figure 1: A two-dimensional lattice.

The number of independent directions in d dimensions is d. For example,

$$d = 2 \quad \Longrightarrow \quad \mu = 1, 2, \tag{5}$$

while

$$d = 3 \quad \Longrightarrow \quad \mu = 1, 2, 3. \tag{6}$$

Thus, in general,

$$x^\mu = n^\mu \Delta, \qquad n^\mu \in \mathbb{Z}^d. \tag{7}$$

The Green function can be represented as a sum over all paths connecting the origin to x:

$$\boxed{G(m_0, x) = \sum_{0 \to x} \mathrm{e}^{-m_0\, \ell(\text{path})}} \tag{8}$$

where $\ell(\text{path})$ denotes the length of the path.

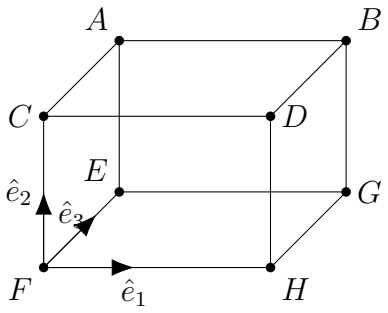

Figure 2: A three-dimensional lattice and its basis vectors.

2. Example

Consider a path from A to C. There are two shortest paths:

$$A \longrightarrow B \longrightarrow C, \tag{9}$$

and

$$A \longrightarrow D \longrightarrow C. \tag{10}$$

The corresponding diagram is

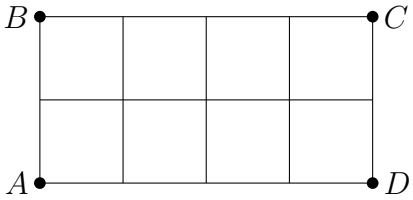

Figure 3: Two shortest paths from A to C.

Therefore,

$$A \to C \equiv \begin{cases} A \to B \to C, \\ A \to D \to C. \end{cases} \tag{11}$$

Each shortest path has two lattice steps, so that

$$\ell(\text{path}) = 2\Delta. \tag{12}$$

Hence the contribution from these two paths is

$$G = \sum_{A \to C} \mathrm{e}^{-m_0 \ell(\text{path})} \tag{13}$$

$$= \mathrm{e}^{-m_0(2\Delta)} + \mathrm{e}^{-m_0(2\Delta)} \tag{14}$$

$$= 2\mathrm{e}^{-2m_0\Delta}. \tag{15}$$

More generally, let $N_j(n)$ denote the number of paths of length j connecting the origin to the lattice point n. Then

$$\boxed{G(m_0, n) = \sum_{j=0}^{\infty} N_j(n)\, \mathrm{e}^{-m_0 j\Delta}} \tag{16}$$

3. Continuum Limit

Since

$$n = \frac{x}{\Delta}, \tag{17}$$

we can write

$$G(x, m_0) = \sum_{j=0}^{\infty} N_j\left(\frac{x}{\Delta}\right) \mathrm{e}^{-m_0 j\Delta}. \tag{18}$$

In the continuum limit,

$$\Delta \to 0, \tag{19}$$

we therefore have

$$G(x, m_0) = \lim_{\Delta\to 0} \sum_{j=0}^{\infty} N_j\left(\frac{x}{\Delta}\right) \mathrm{e}^{-m_0 j\Delta}. \tag{20}$$

The number of paths grows as the lattice becomes finer. Therefore, the mass parameter has to be treated carefully in the continuum limit. In particular, we allow

$$m_0 = m_0(\Delta). \tag{21}$$

The small-Δ expansion is

$$\boxed{m_0(\Delta) = m_0(0) + \left.\frac{dm_0}{d\Delta}\right|_{\Delta=0} \Delta + \mathcal{O}(\Delta^2)}. \tag{22}$$

4. Expansion

We have

$$\mathrm{e}^{-m_0(\Delta) j\Delta} = \mathrm{e}^{-m_0(0) j\Delta} \exp\left[-j\Delta\left(\left.\frac{dm_0}{d\Delta}\right|_{\Delta=0} \Delta + \cdots\right)\right]. \tag{23}$$

Keeping the leading terms,

$$\mathrm{e}^{-m_0(\Delta) j\Delta} \simeq \mathrm{e}^{-m_0(0) j\Delta}\left[1 - j\left.\frac{dm_0}{d\Delta}\right|_{\Delta=0} \Delta + \cdots\right]. \tag{24}$$

Consequently,

$$G(x, m_0) \simeq \sum_{j=0}^{\infty} \mathrm{e}^{-m_0(0) j\Delta} \left[N_j\left(\frac{x}{\Delta}\right) + j N_j\left(\frac{x}{\Delta}\right) \left.\frac{dm_0}{d\Delta}\right|_{\Delta=0} \Delta \right] \tag{25}$$

$$+ \mathcal{O}(\Delta^2). \tag{26}$$

The change in the number of paths can be written schematically as

$$\Delta N_j \simeq x \frac{dN_j}{dn}. \tag{27}$$

Thus the behavior of $N_j(n)$ as $n \to \infty$ becomes important in the continuum limit.

5. Fourier Representation

For a translationally invariant lattice, one may introduce a Fourier representation of the path-counting function. Schematically,

$$\mathrm{e}^{ik_\mu x^\mu} = f(x), \tag{28}$$

and the behavior of $N_j(n)$ for large n determines the continuum behavior of the Green function.

The important point is that as

$$\Delta \to 0, \tag{29}$$

the lattice point

$$n = \frac{x}{\Delta} \tag{30}$$

goes to infinity for fixed nonzero x. Therefore, the asymptotic growth of $N_j(n)$ has to be compensated by the exponential factor in the propagator.

6. Growth of the Number of Paths

Consider the Green function

$$\boxed{G(m_0, x) = \sum_{j=0}^{\infty} N_j\left(\frac{x}{\Delta}\right) \mathrm{e}^{-m_0 j\Delta}}. \tag{31}$$

A path of length j contains j elementary steps. In two dimensions there are four possible nearest-neighbor directions,

$$\pm\hat{e}_1, \qquad \pm\hat{e}_2, \tag{32}$$

so that at each step there are at most

$$2d = 4 \tag{33}$$

possible choices.

In the same way, in four dimensions the number of nearest-neighbor directions is

$$2d = 8. \tag{34}$$

Thus the number of possible paths grows exponentially. For the four-dimensional case,

$$N_{2L}(x) \gtrsim 8^L, \tag{35}$$

and in particular,

$$N_{2L}(0) \gtrsim 8^L. \tag{36}$$

This exponential growth is the source of the divergence of the path sum unless the exponential suppression from the factor $\mathrm{e}^{-m_0 j\Delta}$ is sufficiently strong.

7. Condition for Convergence

Starting from

$$G(m_0, x) = \sum_{j=0}^{\infty} N_j(n)\, \mathrm{e}^{-m_0 j\Delta}, \tag{37}$$

and using the estimate

$$N_{2L}(x) \gtrsim 8^L, \tag{38}$$

we obtain

$$G(m_0, x) \gtrsim \sum_{L=0}^{\infty} 8^L \mathrm{e}^{-m_0\Delta(2L)} \tag{39}$$

$$= \sum_{L=0}^{\infty} \left(8\mathrm{e}^{-2m_0\Delta}\right)^L. \tag{40}$$

Therefore, convergence requires

$$8\mathrm{e}^{-2m_0\Delta} < 1. \tag{41}$$

Equivalently,

$$\mathrm{e}^{-2m_0\Delta} < \frac{1}{8}, \tag{42}$$

which gives

$$-2m_0\Delta < -\ln 8, \tag{43}$$

and hence

$$\boxed{m_0 > \frac{\ln 8}{2\Delta}}. \tag{44}$$

Thus the bare mass parameter must grow appropriately as $\Delta \to 0$ in order to suppress the exponential growth of the number of lattice paths.

8. Path Representation

The sum over paths can be illustrated schematically as follows.

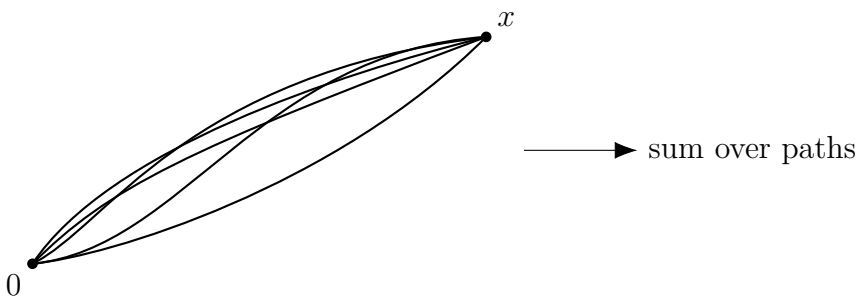

Figure 4: Schematic representation of paths from 0 to x.

Hence the Green function is the weighted sum

$$G(m_0, x) = \sum_{\text{paths } 0 \to x} \mathrm{e}^{-m_0 \ell(\text{path})}. \tag{45}$$

Each path is weighted by its length, and the factor $\mathrm{e}^{-m_0 \ell}$ suppresses long paths.

AI-Assisted Preparation

This document was prepared with the assistance of **ChatGPT (GPT-5.6 Luna)**, an artificial intelligence model developed by **OpenAI**. The assistance included transcription and formatting of the handwritten material, LaTeX typesetting, organization of the equations, and reconstruction of the diagrams using TikZ.

The mathematical content is based on the original submitted material.